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(54) **METHODS AND SYSTEMS FOR
IMPLEMENTING A LOCK-OUT COMMAND
ON LEVER MACHINES**

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(57) **ABSTRACT**

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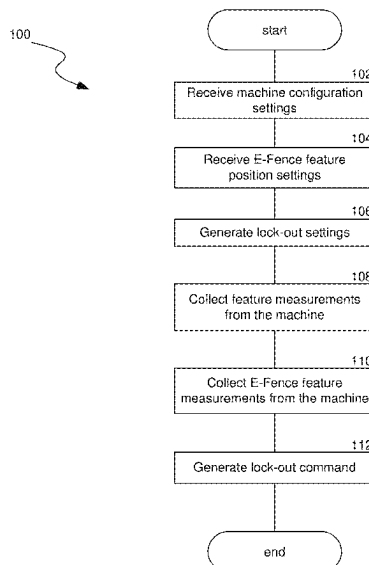
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A technique is directed to methods and systems of an implement lock-out on lever-controlled machines. A lock-out system can monitor the position of an implement and lock-out the implement control(s) when the implement is within a threshold distance to parts of the machine. The lock-out system can generate an implement lock-out to slow, stop, or reduce the force of a hydraulic valve(s) controlling the implement. The lock-out system can use inputs such as electronic fence blade position system data, articulation angles, wheel lean angles, steering angles, ripper positions, mode selection or similar data to determine to generate the implement lock-out. The lock-out system can generate the implement lock-out by a flow supply shutoff to the implement while maintaining pressure to the steering valve. The lock-out system can send visual or audible notifications to alert the operator of the implement's proximity to the machine or of an implement lock-out.

20 Claims, 10 Drawing Sheets



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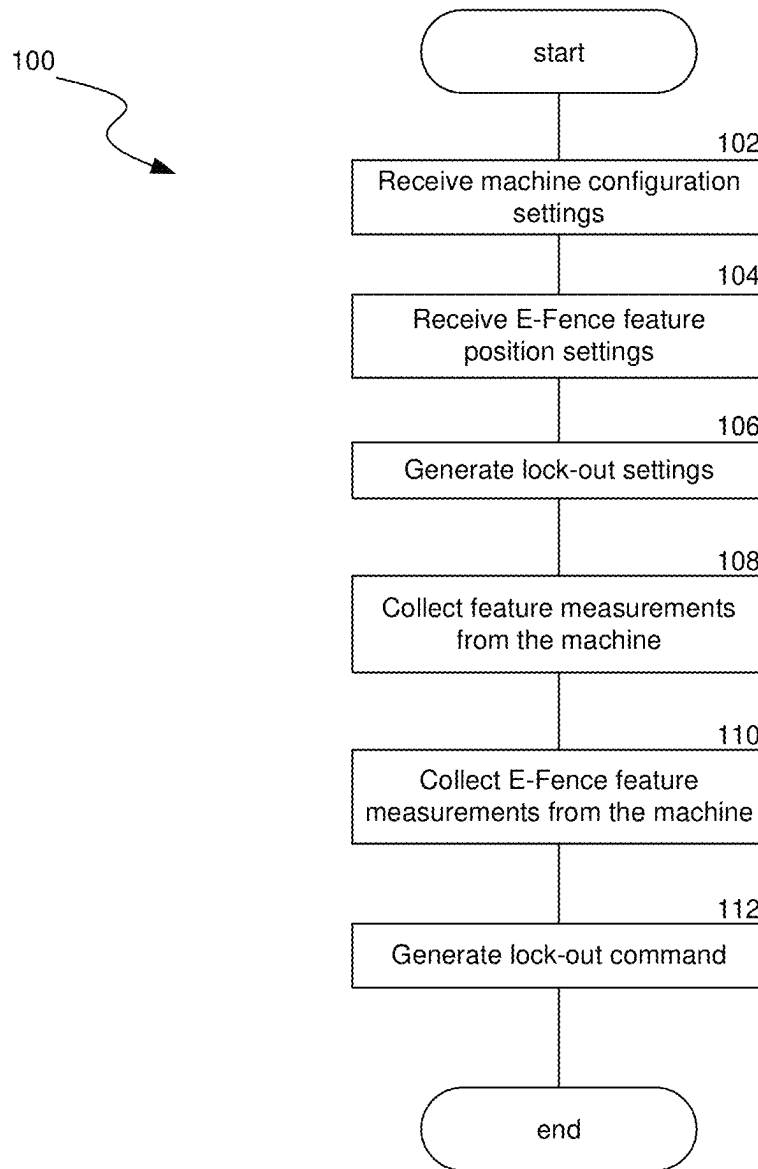
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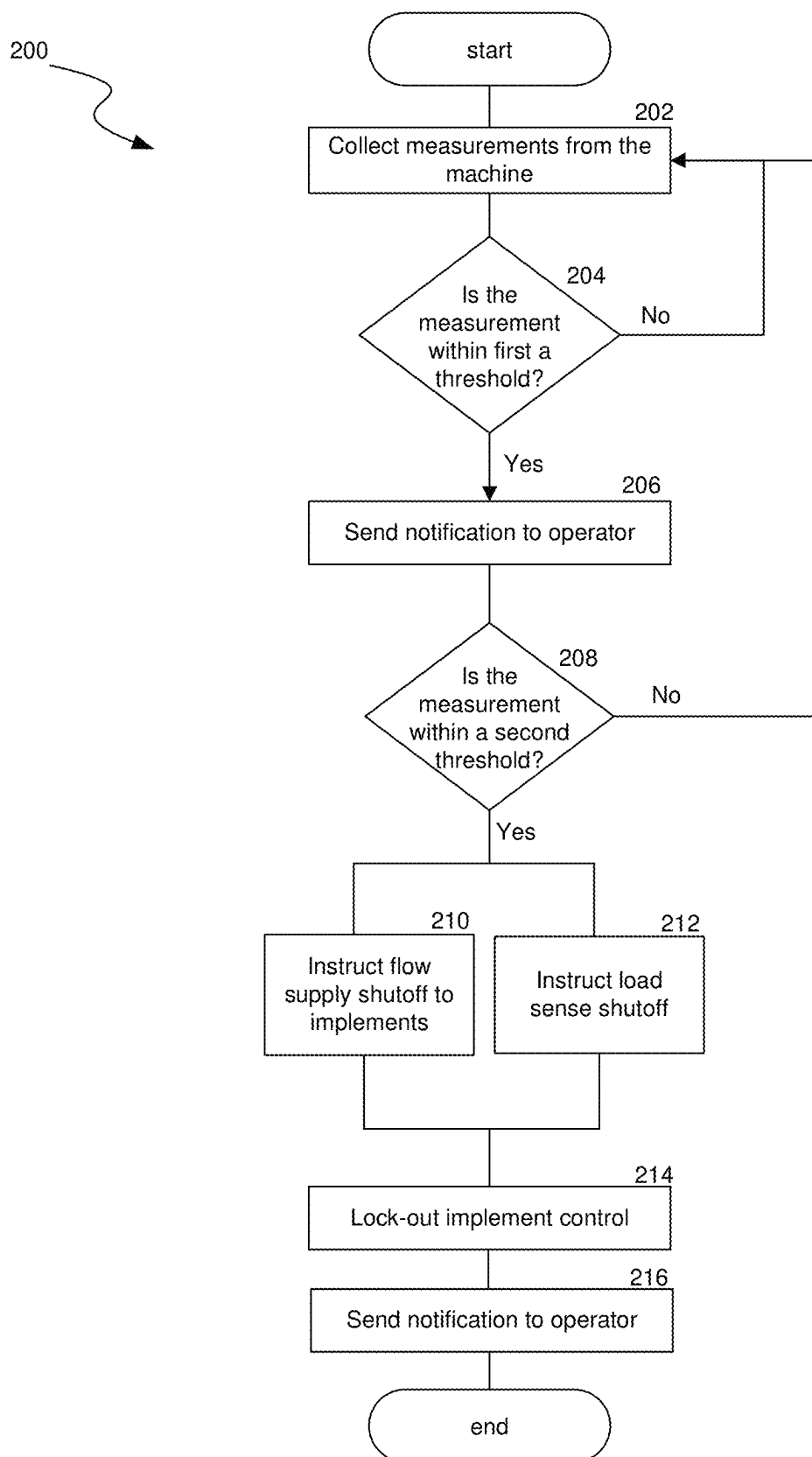
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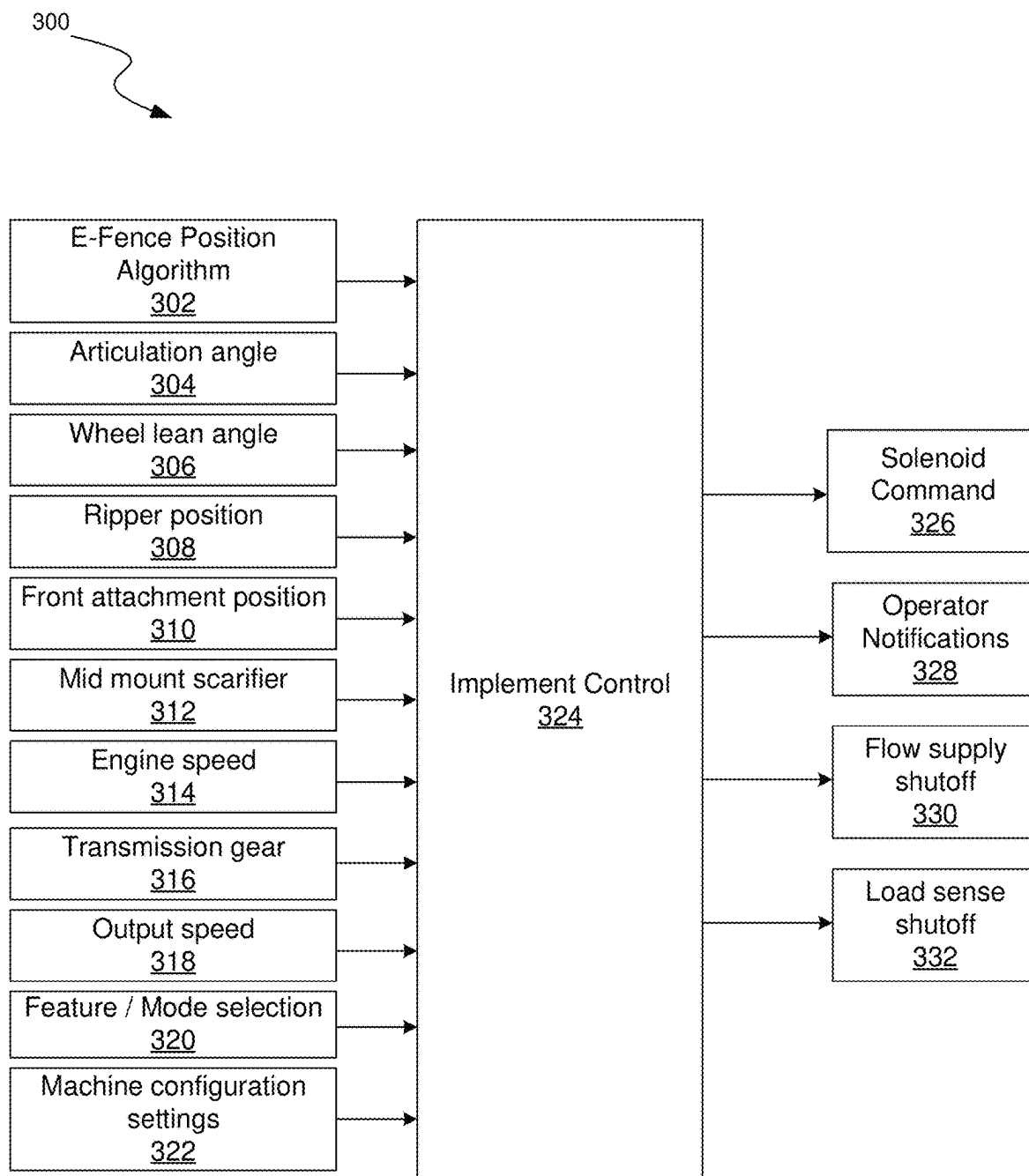
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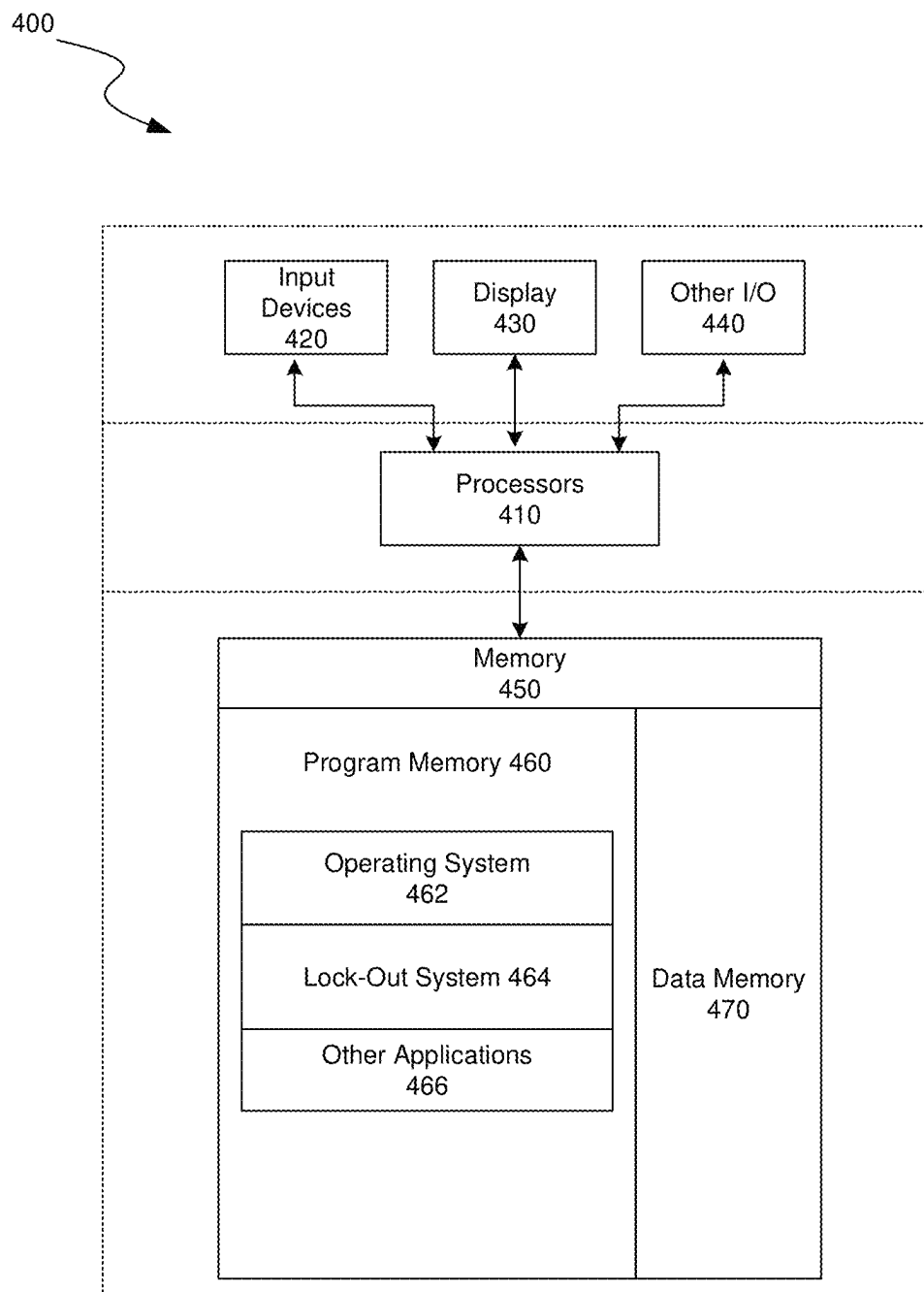
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**FIG. 1**

**FIG. 2**

**FIG. 3**

**FIG. 4**

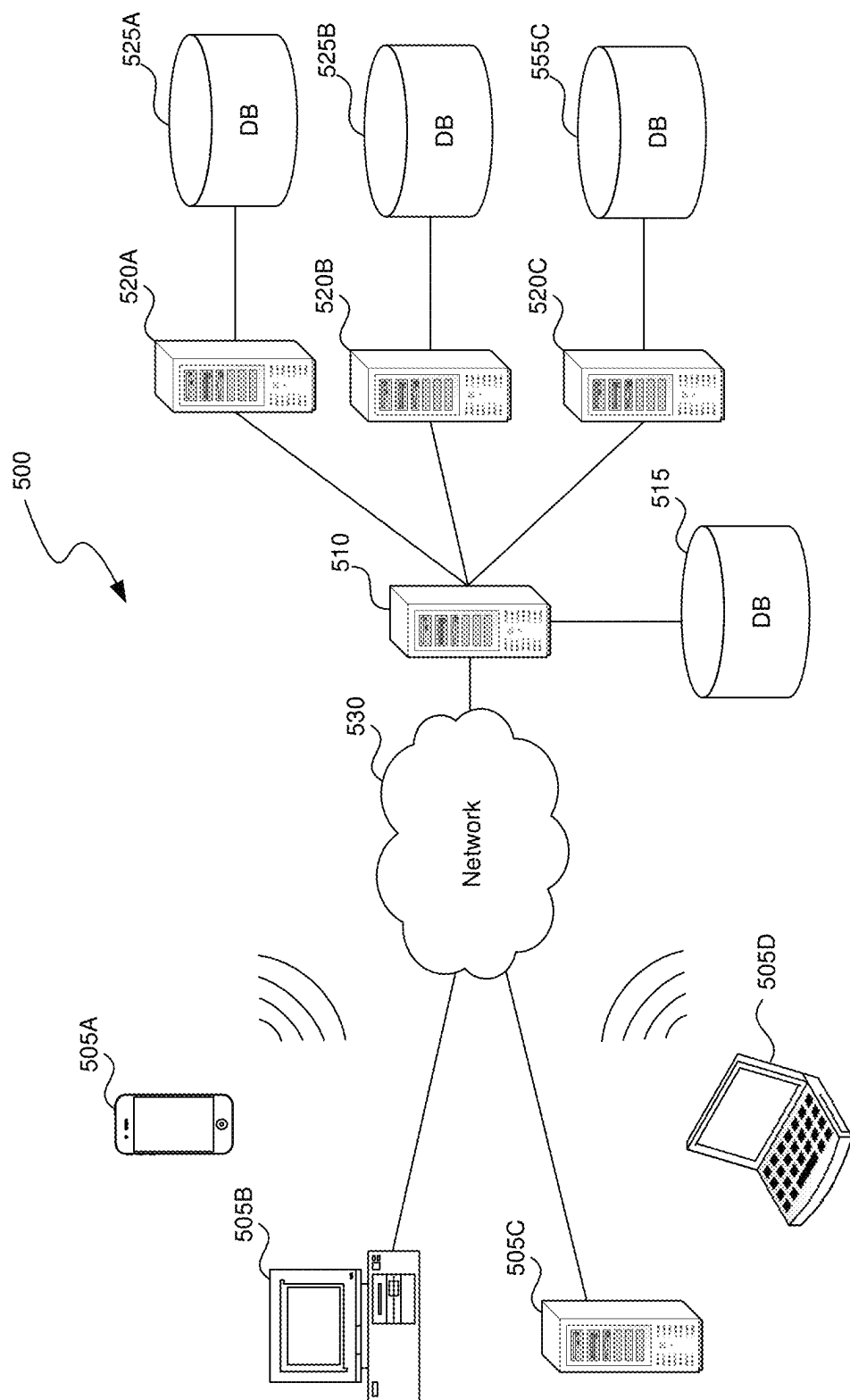
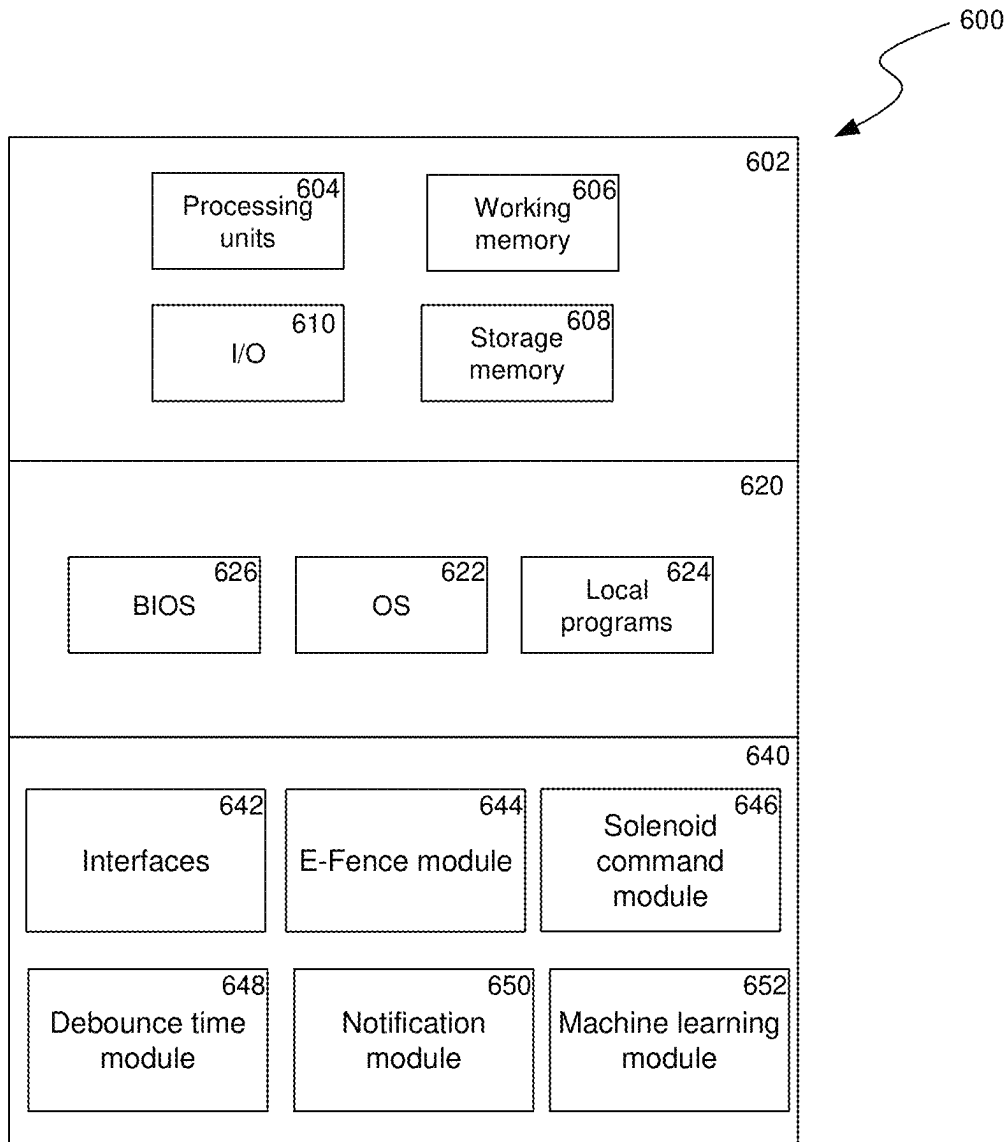


FIG. 5

**FIG. 6**

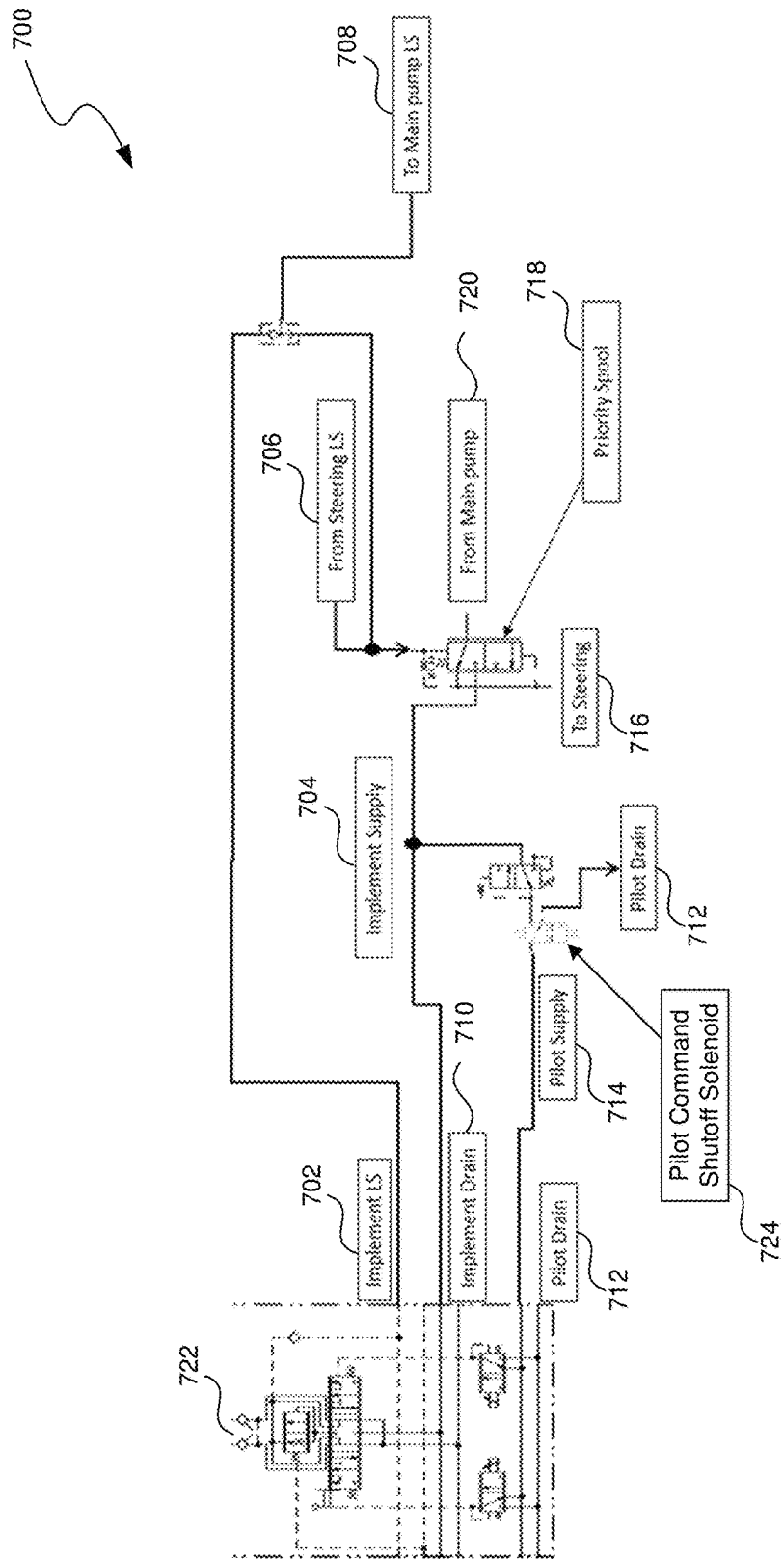


FIG. 7

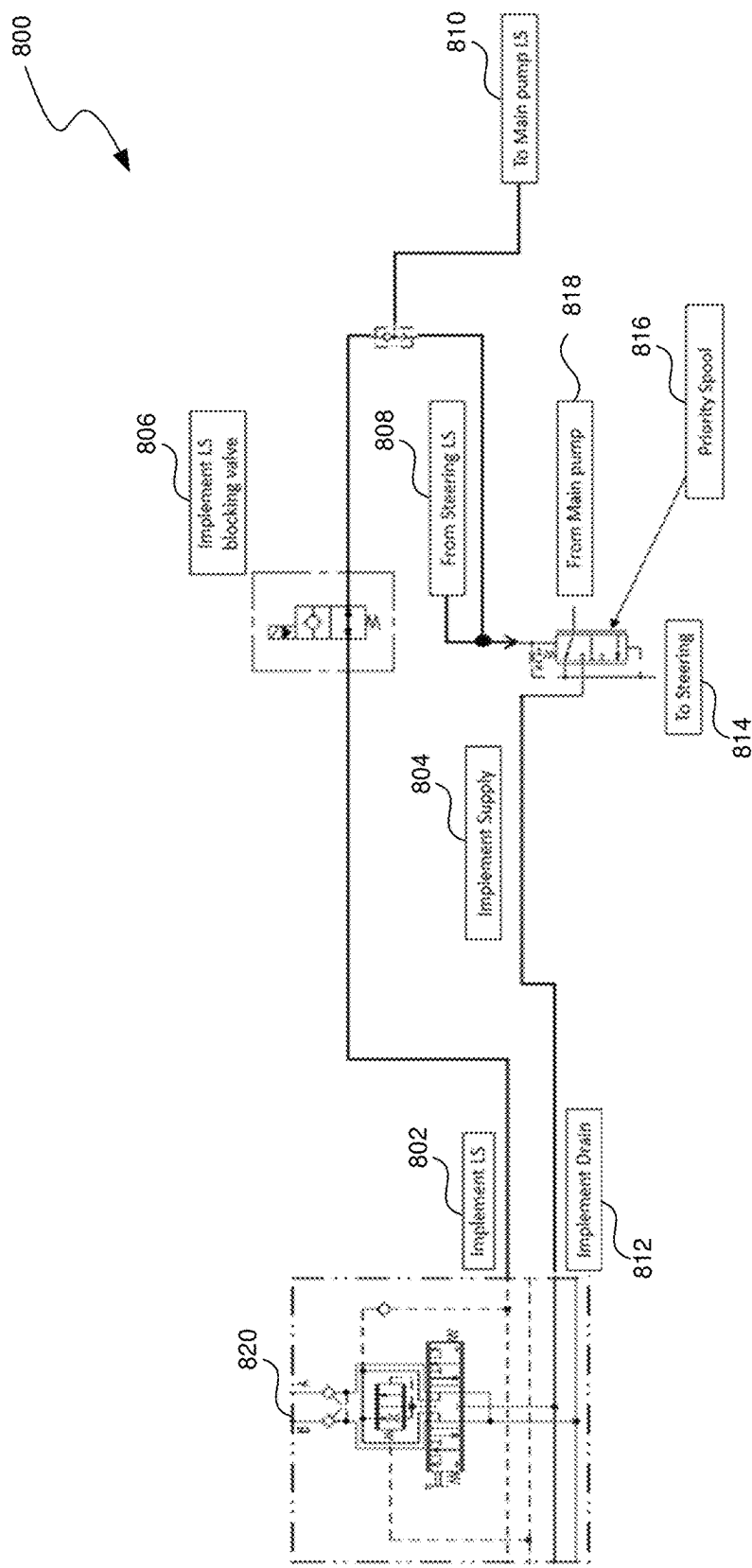


FIG. 8

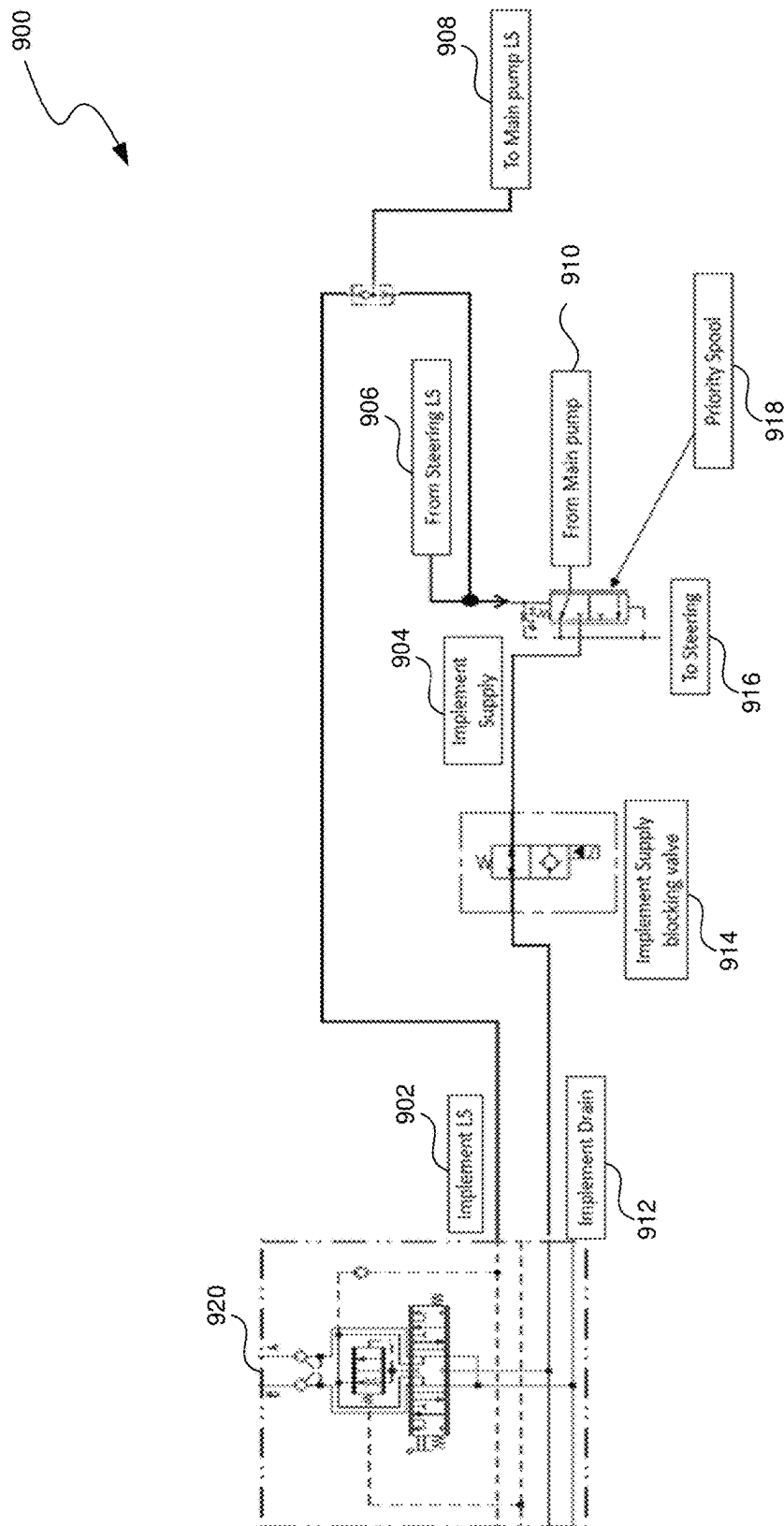


FIG. 9

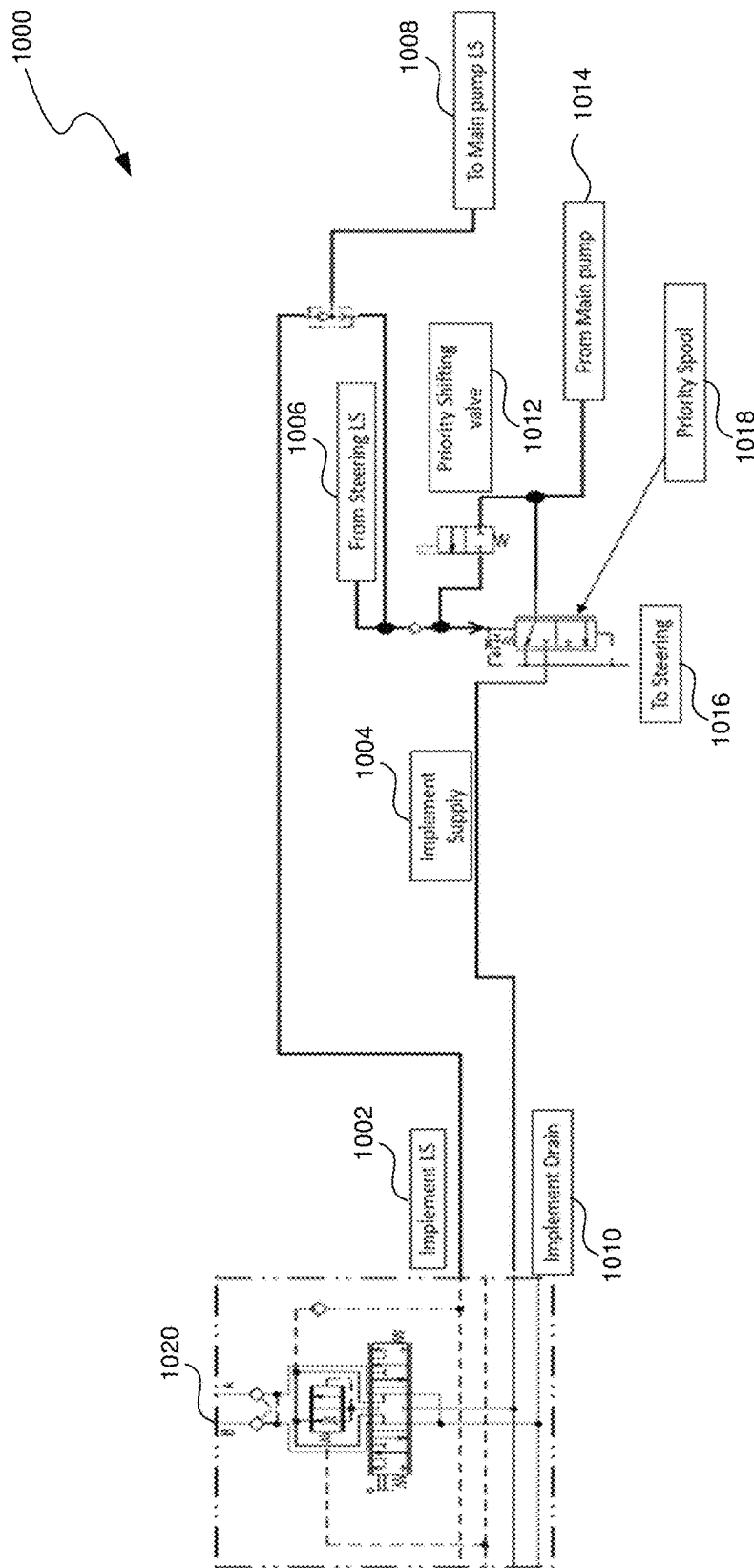


FIG. 10

METHODS AND SYSTEMS FOR IMPLEMENTING A LOCK-OUT COMMAND ON LEVER MACHINES

BACKGROUND

Users may operate machinery such as earthmoving or construction equipment in worksite environments. However, as a user operates the machine, implements to the machine, such as a blade, ripper, or bucket, can potentially cause damage to the machine and/or operator if the operator errs and moves the implement into the machine. Safety for operators and protecting equipment are growing concerns in the industry, and companies have implemented prevention techniques to protect operators and equipment from dangerous events. For example, U.S. Pat. No. 9,238,900B2 describes a method for driving a boom actuator of a front loader which adjusts pivot angles and scoop angles to prevent inadvertent contact between the bucket and the boom. However, this method is only directed to detecting a stop angle to keep the bucket from contacting the boom. Additionally, U.S. Pat. No. 10,815,640B2 describes a method for causing a wheel loader to raise the boom to avoid a collision with the loading target. However, this method is only directed to increasing the engine speed to provide the hydraulic pump with enough power to lift the bucket to a determined height to avoid the collision.

SUMMARY

In some embodiments, a method for implement lock-out on lever-controlled machines, such as earthmoving or construction equipment, includes receiving machine feature measurements for a lever-controlled machine operating at a site and determining whether the machine feature measurements are within a measurement threshold. In response to the machine feature measurements being within the measurement threshold, the method can include generating an implement lock-out of at least one hydraulic implement on the lever-controlled machine, wherein the implement lock-out at least one of slows or stops a flow supply to the hydraulic implement on the lever-controlled machine. The operator of the lever-controlled machine can have a steering capability during the implement lock-out. The method can further include sending, to the operator of the lever-controlled machine, a notification of the implement lock-out and disabling the implement lock-out on the lever-controlled machine after a time threshold.

In some embodiments, a system for implement lock-out on lever-controlled machines, such as earthmoving or construction equipment, include a machine and one or more non-transitory computer-readable media. The system can include receiving machine feature measurements for a lever-controlled machine operating at a site and determining whether the machine feature measurements are within a measurement threshold. In response to the machine feature measurements being within the measurement threshold, the system can generate an implement lock-out of at least one hydraulic implement on the lever-controlled machine, wherein the implement lock-out at least one of slows or stops a flow supply to the hydraulic implement on the lever-controlled machine. The operator of the lever-controlled machine can have a steering capability during the implement lock-out. The system can further include sending, to the operator of the lever-controlled machine, a notification

of the implement lock-out and disabling the implement lock-out on the lever-controlled machine after a time threshold.

In some embodiments, the machine can further include one or more processors; and one or more memory devices having stored thereon instructions that when executed by the one or more processors cause the one or more processors to perform at least one of the following: (i) receive machine feature measurements for a lever-controlled machine operating at a site; (ii) determine the machine feature measurements are within a measurement threshold; (iii) in response to the machine feature measurements being within the measurement threshold, generating an implement lock-out of at least one hydraulic implement on the lever-controlled machine, wherein the implement lock-out at least one of slows or stops a flow supply to the hydraulic implement on the lever-controlled machine, wherein an operator of the lever-controlled machine has a steering capability during the implement lock-out; (iv) send, to the operator of the lever-controlled machine, a notification of the implement lock-out; and (v) disable the implement lock-out on the lever-controlled machine after a time threshold.

In some embodiments, the machine can further include an apparatus with a memory; one or more processors electronically coupled to the memory and configured to: (i) receive machine feature measurements for a lever-controlled machine operating at a site; (ii) determine the machine feature measurements are within a measurement threshold; (iii) in response to the machine feature measurements being within the measurement threshold, generating an implement lock-out of at least one hydraulic implement on the lever-controlled machine, wherein the implement lock-out at least one of slows or stops a flow supply to the hydraulic implement on the lever-controlled machine, wherein an operator of the lever-controlled machine has a steering capability during the implement lock-out; (iv) send, to the operator of the lever-controlled machine, a notification of the implement lock-out; and (v) disable the implement lock-out on the lever-controlled machine after a time threshold.

Other aspects will appear hereinafter. The features described herein can be used separately or together, or in various combinations of one or more of them.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a flow diagram illustrating a process used in some implementations for generating an implement lock-out.

FIG. 2 is a flow diagram illustrating a process used in some implementations for locking out an implement control.

FIG. 3 is a schematic diagram illustrating an example of an implement lock-out system.

FIG. 4 is a block diagram illustrating an overview of devices on which some implementations can operate.

FIG. 5 is a block diagram illustrating an overview of an environment in which some implementations can operate.

FIG. 6 is a block diagram illustrating components which, in some implementations can be used in a system employing the disclosed technology.

FIG. 7 is a schematic diagram illustrating an example of a pilot shutoff command in an implement lock-out system.

FIG. 8 is a schematic diagram illustrating an example of an implement load sense signal shutoff in an implement lock-out system.

FIG. 9 is a schematic diagram illustrating an example of an implement supply shutoff in an implement lock-out system.

FIG. 10 is a schematic diagram illustrating an example of a priority spool control in an implement lock-out system.

The techniques introduced here may be better understood by referring to the following Detailed Description in conjunction with the accompanying drawings, in which like reference numerals indicate identical or functionally similar elements.

DETAILED DESCRIPTION

Aspects of the present disclosure are directed to methods and systems for an implement lock-out on lever-controlled machines such as earthmoving or construction equipment. In some implementations, operators control one or more implements (e.g., blades, rippers, buckets, adapters, augers, bale grabs, brooms, bale spears, brush-cutters, etc.) with lever controls. Note that while this description will refer to a single implement, the singular implicates the plural and multiple implements could be involved. As the operator moves the various levers, the levers activate hydraulic valves that control the machine implement, as there is a direct mechanical connection from the operator input lever to flow control on the implement hydraulic valves. In some cases, the operator can inadvertently move the implement in a way that causes damage to the machine or parts (e.g., wheels, cab, attachments, etc.) of the machine. For example, an operator of a motor grader causes the blade to collide with a tire on the machine.

In an embodiment, a lock-out system monitors the position of the implement and locks-out the implement controls when the implement is within a threshold distance (e.g., any distance, such as 0.5 inches, 1 inch, 6 inches, 12 inches, etc.) to parts of the machine. The lock-out system can generate an implement lock-out to slow, stop, or reduce the force of hydraulic actuator(s) for the implement based on machine control logic, such as independent of operator input. In some cases, the machine control logic is used when a feature is assisting the operator with training or protecting the machine from damage. The machine control logic can use inputs such as electronic fence (E-fence) blade position system, articulation angle, wheel lean angle, steering angle, ripper position, mode selection or similar data to determine to generate the implement lock-out. The lock-out system can generate the implement lock-out by a flow supply shutoff to the hydraulic actuator(s) for the implement or by a load sense shutoff. In a flow supply shutoff to the implement, the steering system still receives a hydraulic flow supply, which permits the operator to maintain steering capability of the machine during the implement lockout.

In some embodiments, the lock-out system can lock-out the implement controls by a load sense shutoff. In a load sense shutoff, the lock-out system will reduce the pressure or flow from the implement hydraulic actuator(s) back to the pump which will reduce the implement pressure to a standby value. For example, during a lock-out, as an implement control is moved (e.g., blade is raised on a machine), the lock-out system will keep the pressure low enough so the hydraulic valve does not move the implement. In some embodiments, the lock-out system can lock-out the implement controls by a solenoid or pilot command shutoff for electro-hydraulic controls. For example, during a lock-out, the lock-out system keeps the spool shifted only to steering so that the solenoid can manually lock-out the mechanical lever controls with an electronic feature.

The lock-out system can send notifications (e.g., visual or audible warnings, such as flashing lights, vibration in the operator's seat, vibration in the control levers or steering instrument, or an alarm) to alert the operator of the implement's proximity to the machine. The notification can alert the operator to the implement lock-out and the duration of the implement lock-out. In some implementations, the operator activates the lock-out command on the implement controls. For example, as the operator navigates the machine at a worksite or along a road, the operator can lock-out the implement controls to prevent the implement from damaging the machine.

Several implementations are discussed below in more detail with reference to the figures. FIG. 1 is a flow diagram illustrating a process 100 used in some implementations for generating an implement lock-out. In an embodiment, process 100 is triggered by any of the machine powering on, a user (e.g., operator) pressing a button on a control device or inputting a command, attaching an implement to a machine, the type of implement on a machine, the type of machine, or process 100 is always operating while the machine is powered on. Examples of suitable machines are, but not limited to, bulldozers, excavators, trenchers, loaders, backhoes, compactors, graders, feller bunchers, graders, wheel tractor scrapers, skid-steer loaders, dump trucks, cranes, telehandlers, pavers, and pile-driving/boring machines.

At step 102, process 100 receives machine configuration settings for the lever-controlled machine in operation. The machine configuration settings can include the tire size, rim size, blade length, attachments, the articulation angle, wheel lean angle, steering angle, ripper position, front attachment position, mid mount scarifier position, engine speed, transmission gear, output speed, and the feature and/or mode selection. The feature and/or mode selection can include machine autopilot or operator assist switched on/off, bound box settings around the machine for protection, or aggressive settings.

At step 104, process 100 receives E-Fence feature position settings. The E-Fence feature position settings indicate the position and orientation of the parts, attachments, and the implement of a machine. Process 100 uses the E-fence position settings to monitor the machine and the implement on the machine while the machine is operating. The E-Fence feature position settings can indicate a detection zone (e.g., protection zone) around the parts of the machine which is created by sensors. For example, the sensors detect if an implement is within a threshold distance to a part of the machine, such as the blade is within a threshold distance to a wheel.

At step 106, process 100 generates implement lock-out parameters for the machine using the E-Fence position settings and/or the machine configuration settings. Process 100 can generate a threshold distance that if the implement is within, an alarm or lock-out is triggered. Process 100 can determine the duration of an implement lock-out based on the E-Fence position settings and/or the machine configuration settings. The E-Fence position settings and/or the machine configuration settings can indicate the duration of the lock-out based on the type of machine, type of implement, or the proximity of the implement to the machine. In an example, if the implement comes within a determined distance (e.g., any distance, such as an inch, foot, etc.) to the machine, the lock-out requires a manual reset by the operator. In another example, if the implement comes within a determined distance to the machine, the implement lock-out remains active for a duration (e.g., 5 sec, 10 sec, etc.) and deactivates after the duration is complete. In some embodi-

ments, the implement lock-out can remain active until the operator acknowledges (e.g., presses a button, flips a switch, etc.) the lock-out.

At step **108**, process **100** collects feature measurements from the machine. Process **100** can collect feature measurements (e.g., tire size, rim size, blade length, attachments, the articulation angle, wheel lean angle, steering angle, ripper position, front attachment position, mid mount scarifier position, engine speed, transmission gear, and/or output speed) from the machine and compare the current measurements to the feature settings. If one or more of the feature measurements are greater by a threshold amount than one or more of the corresponding feature settings, process **100** can generate an implement lock-out.

At step **110**, process **100** collects E-Fence feature measurements from the machine. The E-fence feature measurements can indicate the position and orientation of the implement in relation to the machine. For example, the E-Fence feature measurements indicate the distance from the implement to the machine or machine part, such as the distance from a blade to a tire on the machine. Process **100** can compare the E-fence feature measurements to the machine feature measurements to determine the distance from the implement to the machine or a part of the machine.

At step **112**, process **100** generates the implement lock-out command for the hydraulic valves on lever-controlled machine.

FIG. 2 is a flow diagram illustrating a process **200** used in some implementations for locking out an implement control. In an embodiment, process **200** is triggered by any of the machine powering on, a user (e.g., operator) pressing a button on a control device or inputting a command, attaching an implement to a machine, the type of implement on a machine, the type of machine, or process **200** is always operating while the machine is powered on.

At step **202**, process **200** collects measurements from the machine. The measurements can include the machine feature measurements and the E-Fence feature measurements as describes at steps **108** and **110** of FIG. 1. At step **204**, process **200** determines whether the measurements are within a first threshold. When the measurements are not within the first threshold (e.g., indicating the implement is within a safe operating distance from the machine features), process **200** can continue to collect measurements at step **202**.

At step **206**, process **200** sends a notification to the operator when the measurements are within a first threshold, indicating the orientation and position of the implement in relation to the machine. The first threshold distance can serve as a “warning” zone to alert the operator to the proximity of the implement to a part of the machine. If the operator adjusts the position or orientation of the implement to shift outside the first threshold, process **200** will continue to collect measurements and monitor the position and orientation of the implement.

At step **208**, process **200** determines whether the measurements are within a second threshold. If the measurements are within a second threshold, process **200** can generate a lock-out by instructing either a flow supply shutoff to the implement or a load sense shutoff. At step **210**, process **200** instructs a flow supply shutoff to the implement. A flow supply shutoff to the implement can indicate a priority spool has shifted to steering only. For example, as the operator moves a lever, the direct mechanical connection from the operator input lever to flow control on the implement hydraulic valves is interrupted so that the implement is unable to move. Maintaining steering capability permits the operator to navigate the machine without an implement

causing damage to the machine. For example, during a lock-out, the operator can move the motor grader without the blade contacting a tire on the motor grader. A flow supply shutoff can close off the machine’s main flow going to the implement. The main flow can go to the priority spool and whatever flow the steering does not use is available to the implement(s). For example, the flow supply shutoff would block the flow with a solenoid valve or a signal that shifts the priority spool to only allow steering to receive flow.

Example **900** of FIG. 9 can illustrate an implement supply shutoff. The flow from implement valve **920** can travel through the implement load sense (LS) **902** to the main pump LS **908**. The priority spool **918** can receive inputs from the main pump **910** and implement supply **904** and output to the steering **916**. The flow from pump **910** can travel through the implement drain **912** to implement valve **920** through implement supply **904**. The implement supply blocking valve **914** can close off the main flow going to the implement valve **920**.

Example **1000** of FIG. 10 can illustrate a priority spool control. The flow from implement valve **1020** can travel through the implement load sense (LS) **1002** to the main pump LS **1008**. The priority spool **1018** can receive inputs from the main pump **1014** and implement supply **1004** and output to the steering **1016**. The flow from pump **1014** can travel to implement valve **1020** through implement supply **1004**. The priority shifting valve **1012** can close off the flow going to the implements and shifts the priority spool to only allow steering to receive flow.

At step **212**, process **200** instructs a load sense shutoff. In a load sense shutoff, process **200** reduces the pressure or flow from the implement back to the pump which will reduce the implement pressure to a standby value. For example, during a lock-out, as an implement valve is moved (e.g., blade is raised on a machine), the lock-out system will keep the pressure low enough so the hydraulic valve does not move the implement. In a load sense shutoff, load sense or signal flow is sent from the implement back to the main pump. The machine can use variable piston pumps which take a signal and determine, based on a margin pressure, to increase flow until the margin pressure is met. The load sense shutoff can block or divert the load sense from going back to the controller on the pump. At step **214**, process **200** locks-out the implement controls.

Example **800** of FIG. 8 illustrates an implement load sense (LS) signal shutoff. The flow from implement valve **820** can travel through the implement load sense (LS) **802** to the main pump LS **810**. Implement LS blocking valve **806** reduces the pressure or flow from the implement back to the pump **818**. The priority spool **816** can receive inputs from the main pump **818** and implement supply **804** and output to the steering **814**. The flow from pump **818** can travel to implement valve **820** through implement supply **804**.

At step **216**, process **200** sends a notification to the operator regarding the lock-out. The notification can be audible and/or visual to alert the operator to the implement lock-out. The notification can last the duration of the lock-out. In some embodiments, the notification indicates the duration or lock-out and provides the operator with instructions. The instructions can include steps the operator can perform to deactivate the lock-out or the duration of the lock-out.

FIG. 3 is a schematic diagram illustrating example **300** of an implement lock-out system. The implement control **324** (e.g., running a control algorithm) can receive data inputs from the machine, such as one or more of E-Fence position algorithm **302**, articulation angle **304**, wheel lean angle **306**,

ripper position **308**, front attachment position **310** (e.g., blade, plow, loader, etc.), mid-mount scarifier **312**, engine speed **314**, transmission gear **316**, output speed **318**, feature/mode selection **320**, and machine configuration settings **322**. Implement control **324** can process the data inputs and generates outputs. The outputs can include one or more of a solenoid command **326**, operator notifications **328**, a flow supply shutoff **330**, and a load sense shutoff **332**.

FIG. **4** is a block diagram illustrating an overview of devices on which some implementations of the disclosed technology can operate. The devices can comprise hardware components of a device **400** that manages entitlements within a real-time telemetry system. Device **400** can include one or more input devices **420** that provide input to the processor(s) **410** (e.g. CPU(s), GPU(s), HPU(s), etc.), notifying it of actions. The actions can be mediated by a hardware controller that interprets the signals received from the input device and communicates the information to the processor(s) **410** using a communication protocol. Input devices **420** may include, for example, a mouse, a keyboard, a touchscreen, an infrared sensor, a touchpad, a wearable input device, a camera- or image-based input device, a microphone, and/or other user input devices.

Processors **410** can be a single processing unit or multiple processing units in a device or distributed across multiple devices. Processors **410** can be coupled to other hardware devices, for example, with the use of a bus, such as a PCI bus or SCSI bus. The processors **410** can communicate with a hardware controller for devices, such as for a display **430**. Display **430** can be used to display text and graphics. In some implementations, display **430** provides graphical and textual visual feedback to a user. In some implementations, display **430** includes the input device as part of the display, such as when the input device is a touchscreen or is equipped with an eye direction monitoring system. In some implementations, the display is separate from the input device. Examples of display devices are: an LCD display screen, an LED display screen, a projected, holographic, or augmented reality display (such as a heads-up display device or a head-mounted device), and so on. Other I/O devices **440** can also be coupled to the processor, such as a network card, video card, audio card, USB, Firewire or other external device, camera, printer, speakers, CD-ROM drive, DVD drive, disk drive, or Blu-Ray device.

In some implementations, the device **400** also includes a communication device capable of communicating wirelessly or wire-based with a network node. The communication device can communicate with another device or a server through a network using, for example, TCP/IP protocols. Device **400** can utilize the communication device to distribute operations across multiple network devices.

The processors **410** can have access to a memory **450** in a device or distributed across multiple devices. A memory includes one or more of various hardware devices for volatile and non-volatile storage, and can include both read-only and writable memory. For example, a memory can comprise random access memory (RAM), various caches, CPU registers, read-only memory (ROM), and writable non-volatile memory, such as flash memory, hard drives, floppy disks, CDs, DVDs, magnetic storage devices, tape drives, and so forth. A memory is not a propagating signal divorced from underlying hardware; a memory is thus non-transitory. Memory **450** can include program memory **460** that stores programs and software, such as an operating system **462**, Lock-Out system **464**, and other application programs **466**. Memory **450** can also include data memory **470**, entitlement data, user data, retrieval data, management

data, authorization token data, configuration data, settings, user options or preferences, etc., which can be provided to the program memory **460** or any element of the device **400**.

Some implementations can be operational with numerous other computing system environments or configurations. Examples of computing systems, environments, and/or configurations that may be suitable for use with the technology include, but are not limited to, personal computers, server computers, handheld or laptop devices, cellular telephones, wearable electronics, gaming consoles, tablet devices, multiprocessor systems, microprocessor-based systems, set-top boxes, programmable consumer electronics, network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, or the like.

FIG. **5** is a block diagram illustrating an overview of an environment **500** in which some implementations of the disclosed technology can operate. Environment **500** can include one or more client computing devices **505A-D**, examples of which can include device **400**. Client computing devices **505** can operate in a networked environment using logical connections through network **530** to one or more remote computers, such as a server computing device **510**.

In some implementations, server **510** can be an edge server which receives client requests and coordinates fulfillment of those requests through other servers, such as servers **520A-C**. Server computing devices **510** and **520** can comprise computing systems, such as device **400**. Though each server computing device **510** and **520** is displayed logically as a single server, server computing devices can each be a distributed computing environment encompassing multiple computing devices located at the same or at geographically disparate physical locations. In some implementations, each server **520** corresponds to a group of servers.

Client computing devices **505** and server computing devices **510** and **520** can each act as a server or client to other server/client devices. Server **510** can connect to a database **515**. Servers **520A-C** can each connect to a corresponding database **525A-C**. As discussed above, each server **520** can correspond to a group of servers, and each of these servers can share a database or can have their own database. Databases **515** and **525** can warehouse (e.g. store) information such as implement data, machine data, machine data, sensor data, notification data, measurement, and alert data. Though databases **515** and **525** are displayed logically as single units, databases **515** and **525** can each be a distributed computing environment encompassing multiple computing devices, can be located within their corresponding server, or can be located at the same or at geographically disparate physical locations.

Network **530** can be a local area network (LAN) or a wide area network (WAN), but can also be other wired or wireless networks. Network **530** may be the Internet or some other public or private network. Client computing devices **505** can be connected to network **530** through a network interface, such as by wired or wireless communication. While the connections between server **510** and servers **520** are shown as separate connections, these connections can be any kind of local, wide area, wired, or wireless network, including network **530** or a separate public or private network.

FIG. **6** is a block diagram illustrating components **600** which, in some implementations, can be used in a system employing the disclosed technology. The components **600** include hardware **602**, general software **620**, and specialized components **640**. As discussed above, a system implementing the disclosed technology can use various hardware

including processing units **604** (e.g. CPUs, GPUs, APUs, etc.), working memory **606**, storage memory **608** (local storage or as an interface to remote storage, such as storage **515** or **525**), and input and output devices **610**. In various implementations, storage memory **608** can be one or more of: local devices, interfaces to remote storage devices, or combinations thereof. For example, storage memory **608** can be a set of one or more hard drives (e.g. a redundant array of independent disks (RAID)) accessible through a system bus or can be a cloud storage provider or other network storage accessible via one or more communications networks (e.g. a network accessible storage (NAS) device, such as storage **515** or storage provided through another server **520**). Components **600** can be implemented in a client computing device such as client computing devices **505** or on a server computing device, such as server computing device **510** or **520**.

General software **620** can include various applications including an operating system **622**, local programs **624**, and a basic input output system (BIOS) **626**. Specialized components **640** can be subcomponents of a general software application **620**, such as local programs **624**. Specialized components **640** can include electronic fence (E-Fence) module **644**, solenoid command module **646**, debounce time module **648**, notification module **650**, machine learning module **652**, and components which can be used for providing user interfaces, transferring data, and controlling the specialized components, such as interfaces **642**. In some implementations, components **600** can be in a computing system that is distributed across multiple computing devices or can be an interface to a server-based application executing one or more of specialized components **640**. Although depicted as separate components, specialized components **640** may be logical or other nonphysical differentiations of functions and/or may be submodules or code-blocks of one or more applications.

In some embodiments, the E-Fence module **644** is configured to monitor the position of an implement (e.g., blade, ripper, bucket, etc.) to detect the orientation of the implement in relation to the machine. The E-Fence module **644** can determine a threshold distance that the implement is to remain from parts of the machine. For example, the E-Fence module **644** can set the threshold distance to 6 inches (or any distance) and monitor the implement to identify if the implement enters the threshold distance. The E-Fence module **644** can monitor the parts of a machine to determine if an implement or objects (e.g., rocks, vehicles, etc.) are within a threshold distance to the machine. For example, the E-Fence module **644** defines detection zones around the machine or parts of the machine to determine if an implement or object is within a distance close enough to cause damage to the machine.

In some embodiments, the solenoid command module **646** is configured to monitor and control the hydraulic flow to the steering or implement. When a lock-out is generated, the solenoid command module **646** can lock-out the lever controls so that the operator is unable to adjust an implement. The solenoid command module **646** can maintain hydraulic flow to steering so that the operator is able to navigate the machine during the lock-out. In some implementations, the solenoid command module **646** performs similarly to a pilot command shutoff for electro-hydraulic controls.

Example **700** of FIG. **7** illustrates a pilot command shutoff for electro-hydraulic controls. The flow from implement valve **722** can travel through the implement load sense (LS) **702** to the main pump LS **708**. The priority spool **718** can

receive inputs from the main pump **720** and implement supply **704** and output to the steering **716**. The flow from pump **720** can travel to implement valve **722** through implement supply **704**. The pilot command shutoff solenoid **724** can stop the pilot supply flow **714** and keep implement valve **722** from EH spool shifting.

In some embodiments, the debounce time module **648** is configured to generate a time threshold (e.g., any amount of time, such as seconds, minutes, etc.) for the implement lock-out. The debounce time module **648** can determine the duration of the implement lock-out based on the type of implement or the position of the implement in relation to the machine. For example, if the blade of the motor grader is within a threshold distance to a wheel, the duration of the lock-out of blade movement is long enough to allow the operator to move the machine without damaging the wheel with the implement.

In some embodiments, the notification module **650** is configured to generate and send a notification (e.g., visual or audible warning) to the operator. The notification module **650** can send a notification to the operator when an implement is within a threshold distance to the machine or part of the machine. For example, as an implement enters into a detection zone, the notification module **650** can alert the operator to adjust the implement before it damages the machine. The notification module **650** can send a notification to the operator to alert the operator of an implement lock-out. The notification can indicate the duration of the lock-out to the operator. For example, lights in the cab of the machine can blink during the lock-out and the frequency of blinking can correspond to the duration of the lock-out.

In some embodiments, the machine learning module **652** is configured to analyze the orientation of an implement on a machine and generate an implement lock-out to prevent the implement from causing damage to the machine. The machine learning module **652** may be configured to identify when to generate an implement lock-out command based on at least one machine-learning algorithm trained on at least one dataset of implement lock-out commands. The at least one machine-learning algorithms (and models) may be stored locally at databases and/or externally at databases. Machine equipment devices may be equipped to access these machine learning algorithms and intelligently identify when to generate an implement lock-out based on at least one machine-learning model that is trained on a dataset of identified lock-out events.

As described herein, a machine-learning (ML) model may refer to a predictive or statistical utility or program that may be used to determine a probability distribution over one or more-character sequences, classes, objects, result sets or events, and/or to predict a response value from one or more predictors. A model may be based on, or incorporate, one or more rule sets, machine learning, a neural network, or the like. In examples, the ML models may be located on the client device, service device, a network appliance (e.g., a firewall, a router, etc.), or some combination thereof. The ML models may process implement lock-out event databases and other data stores to determine when to generate an implement lock-out based on the orientation and position of the implement on a machine. Determining when to generate an implement lock-out may comprise identifying the distance from the machine to the implement and locking out the implement before it causes damage to the machine. Based on an aggregation of data from translated item name databases and platforms, and other user data stores, at least one ML model may be trained and subsequently deployed to automatically identify the position or orientation of the imple-

ment and generate a lock-out command before the implement damages the machine. The trained ML model may be deployed to one or more devices. As a specific example, an instance of a trained ML model may be deployed to a server device and to a client device which communicate with a machine. The ML model deployed to a server device may be configured to be used by the client device when, for example, the client device is connected to the Internet. Conversely, the ML model deployed to a client device may be configured to be used by the client device when, for example, the client device is not connected to the Internet. In some instances, a client device may not be connected to the Internet but still configured to receive satellite signals with item information, such as specific implement information based on the type of implement and machine. In such examples, the ML model may be locally cached by the client device.

Those skilled in the art will appreciate that the components illustrated in FIGS. 4-6 described above, and in each of the flow diagrams discussed below, may be altered in a variety of ways. For example, the order of the logic may be rearranged, substeps may be performed in parallel, illustrated logic may be omitted, other logic may be included, etc. In some implementations, one or more of the components described above can execute one or more of the processes described below.

INDUSTRIAL APPLICABILITY

The systems and methods described herein can implement an implement lock-out on a lever-controlled machine. The lock-out system monitors the position and orientation of the implement on a machine and locks-out the implement controls when the implement is within a threshold distance to parts of the machine. The lock-out system can generate an implement lock-out to slow, stop, or reduce the force of an hydraulic implement based on machine control logic, independent of operator input. The machine control logic can use inputs such as electronic fence (E-fence) blade position measurements, articulation angle measurements, wheel lean angle measurements, steering angle measurements, ripper position measurements, mode selection or similar data to determine when to generate the implement lock-out. The lock-out system can generate the implement lock-out by a flow supply shutoff to the implement or by a load sense shutoff. In a flow supply shutoff to the implement, the steering system still receives a flow supply, which permits the operator to maintain steering capability of the machine during the implement lockout. In a load sense shutoff, the lock-out system will reduce the pressure or flow from the implement back to the pump which will reduce the implement pressure to a standby value. In some embodiments, the lock-out system can lock-out the implement controls by a solenoid or pilot command shutoff for electro-hydraulic controls (e.g., electro-hydraulic over lever controls). For example, during a lock-out, the lock-out system keeps the spool shifted only to permit steering so that the solenoid can manually lock-out the mechanical lever controls with an electronic feature. The present systems and methods can send notifications (e.g., visual or audible) to alert the operator to the implement's proximity to the machine. The notifications can alert the operator to the implement lock-out and the duration of the implement lock-out. In some implementations, the operator activates the lock-out command on the implement controls. The present systems and methods can be implemented to manage and control multiple indus-

trial machines, vehicles and/or other suitable devices such as mining machines, trucks, corporate fleets, etc.

Several implementations of the disclosed technology are described above in reference to the figures. The computing devices on which the described technology may be implemented can include one or more central processing units, memory, input devices (e.g., keyboard and pointing devices), output devices (e.g., display devices), storage devices (e.g., disk drives), and network devices (e.g., network interfaces). The memory and storage devices are computer-readable storage media that can store instructions that implement at least portions of the described technology. In addition, the data structures and message structures can be stored or transmitted via a data transmission medium, such as a signal on a communications link. Various communications links can be used, such as the Internet, a local area network, a wide area network, or a point-to-point dial-up connection. Thus, computer-readable media can comprise computer-readable storage media (e.g., "non-transitory" media) and computer-readable transmission media.

Reference in this specification to "implementations" (e.g. "some implementations," "various implementations," "one implementation," "an implementation," etc.) means that a particular feature, structure, or characteristic described in connection with the implementation is included in at least one implementation of the disclosure. The appearances of these phrases in various places in the specification are not necessarily all referring to the same implementation, nor are separate or alternative implementations mutually exclusive of other implementations. Moreover, various features are described which may be exhibited by some implementations and not by others. Similarly, various requirements are described which may be requirements for some implementations but not for other implementations.

As used herein, being above a threshold means that a value for an item under comparison is above a specified other value, that an item under comparison is among a certain specified number of items with the largest value, or that an item under comparison has a value within a specified top percentage value. As used herein, being below a threshold means that a value for an item under comparison is below a specified other value, that an item under comparison is among a certain specified number of items with the smallest value, or that an item under comparison has a value within a specified bottom percentage value. As used herein, being within a threshold means that a value for an item under comparison is between two specified other values, that an item under comparison is among a middle-specified number of items, or that an item under comparison has a value within a middle-specified percentage range. Relative terms, such as high or unimportant, when not otherwise defined, can be understood as assigning a value and determining how that value compares to an established threshold. For example, the phrase "selecting a fast connection" can be understood to mean selecting a connection that has a value assigned corresponding to its connection speed that is above a threshold.

Unless specifically excluded, the use of the singular above to describe a component, structure, or operation does not exclude the use of plural such components, structures, or operations.

As used herein, the word "or" refers to any possible permutation of a set of items. For example, the phrase "A, B, or C" refers to at least one of A, B, C, or any combination thereof, such as any of: A; B; C; A and B; A and C; B and C; A, B, and C; or multiple of any item such as A and A; B, and C; A, A, B, C, and C; etc.

13

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Specific embodiments and implementations have been described herein for purposes of illustration, but various modifications can be made without deviating from the scope of the embodiments and implementations. The specific features and acts described above are disclosed as example forms of implementing the claims that follow. Accordingly, the embodiments and implementations are not limited except as by the appended claims.

Any patents, patent applications, and other references noted above are incorporated herein by reference. Aspects can be modified, if necessary, to employ the systems, functions, and concepts of the various references described above to provide yet further implementations. If statements or subject matter in a document incorporated by reference conflicts with statements or subject matter of this application, then this application shall control.

We claim:

1. A computing system comprising:
one or more processors; and
one or more memories storing instructions that, when executed by the one or more processors, cause the computing system to perform a process comprising:
receiving machine feature measurements for a lever-controlled machine operating at a site;
determining the machine feature measurements are within a measurement threshold;
in response to the machine feature measurements being within the measurement threshold, generating an implement lock-out of at least one hydraulic implement on the lever-controlled machine by:
maintaining a first flow supply to a steering component of the lever-controlled machine to preserve a steering capability during the implement lock-out, and
stopping a second flow supply to the at least one hydraulic implement on the lever-controlled machine to prevent adjustment of the at least one hydraulic implement during the implement lock-out;
sending, to the lever-controlled machine, a notification of the implement lock-out; and
disabling the implement lock-out on the lever-controlled machine after a time threshold.
2. The computing system of claim 1, the process further comprising:
determining the machine feature measurements are within a warning measurement threshold; and
in response to the machine feature measurements being within the warning measurement threshold, sending, to the lever-controlled machine, a warning notification; wherein the warning measurement threshold has a value such that the warning notification is sent before the generating of the implement lock-out.
3. The computing system of claim 1, the process further comprising:
receiving electronic fence feature position settings for the lever-controlled machine; and
determining the measurement threshold based on the electronic fence feature position settings.
4. The computing system of claim 1, wherein the implement lock-out deactivates at least one hydraulic valve that controls the at least one hydraulic implement.

14

5. The computing system of claim 1, wherein the implement lock-out includes a load sense shutoff.

6. The computing system of claim 1, the process further comprising:

determining the measurement threshold based on input data collected from the lever-controlled machine, wherein the input data includes at least one of electronic fence blade position measurements, articulation angle measurements, wheel lean angle measurements, steering angle measurements, ripper position measurements, mode selection, or machine configuration settings.

7. The computing system of claim 1, the process further comprising:

determining a duration of the implement lock-out based on the received machine feature measurements.

8. An apparatus comprising:

a memory;

one or more processors electronically coupled to the memory and configured for:

receiving machine feature measurements for a lever-controlled machine operating at a site;

determining the machine feature measurements are within a measurement threshold;

in response to the machine feature measurements being within the measurement threshold, generating an implement lock-out of at least one hydraulic implement on the lever-controlled machine by:

maintaining a first flow supply to a steering component of the lever-controlled machine to preserve a steering capability during the implement lock-out, and

stopping a second flow supply to the at least one hydraulic implement on the lever-controlled machine to prevent adjustment of the at least one hydraulic implement during the implement lock-out;

sending, to the lever-controlled machine, a notification of the implement lock-out; and

disabling the implement lock-out on the lever-controlled machine after a time threshold.

9. The apparatus of claim 8, the one or more processors are further configured for:

determining the machine feature measurements are within a warning measurement threshold; and

in response to the machine feature measurements being within the warning measurement threshold, sending, to the lever-controlled machine, a warning notification;

wherein the warning measurement threshold has a value such that the warning notification is sent before the generating of the implement lock-out.

10. The apparatus of claim 8, wherein the one or more processors are further configured for:

receiving electronic fence feature position settings for the lever-controlled machine; and

determining the measurement threshold based on the electronic fence feature position settings.

11. The apparatus of claim 8, wherein the implement lock-out deactivates at least one hydraulic valve that controls the at least one hydraulic implement.

12. The apparatus of claim 8, wherein the implement lock-out includes a load sense shutoff.

13. The apparatus of claim 8, wherein the one or more processors are further configured for:

determining the measurement threshold based on input data collected from the lever-controlled machine,

15

wherein the input data includes at least one of electronic fence blade position measurements, articulation angle measurements, wheel lean angle measurements, steering angle measurements, ripper position measurements, mode selection, or machine configuration settings.

14. The apparatus of claim **8**, wherein the one or more processors are further configured for:

determining a duration of the implement lock-out based on the received machine feature measurements.

15. A method comprising:

receiving machine feature measurements for a lever-controlled machine operating at a site;

determining the machine feature measurements are within a measurement threshold;

in response to the machine feature measurements being within the measurement threshold, generating an implement lock-out of at least one hydraulic implement on the lever-controlled machine by:

maintaining a first flow supply to a steering component of the lever-controlled machine to preserve a steering capability during the implement lock-out, and

stopping a second flow supply to the at least one hydraulic implement on the lever-controlled machine to prevent adjustment of the at least one hydraulic implement during the implement lock-out;

sending, to the lever-controlled machine, a notification of the implement lock-out; and

disabling the implement lock-out on the lever-controlled machine after a time threshold.

16

16. The method of claim **15**, further comprising:

determining the machine feature measurements are within a warning measurement threshold; and

in response to the machine feature measurements being within the warning measurement threshold, sending, to the lever-controlled machine, a warning notification;

wherein the warning measurement threshold has a value such that the warning notification is sent before the generating of the implement lock-out.

17. The method of claim **15**, further comprising:

receiving electronic fence feature position settings for the lever-controlled machine; and

determining the measurement threshold based on the electronic fence feature position settings.

18. The method of claim **15**, wherein the implement lock-out deactivates at least one hydraulic valve that controls the at least one hydraulic implement.

19. The method of claim **15**, wherein the implement lock-out includes a load sense shutoff.

20. The method of claim **15**, further comprising:

determining the measurement threshold based on input data collected from the lever-controlled machine,

wherein the input data includes at least one of electronic fence blade position measurements, articulation angle measurements, wheel lean angle measurements, steering angle measurements, ripper position measurements, mode selection, or machine configuration settings.

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